RANGASWAMY A

Email | +91 8296764997 | Portfolio | GitHub | LinkedIn

SUMMARY

Aspiring Machine Learning Engineer pursuing a B.E. in Artificial Intelligence and Machine Learning, with hands-on experience developing and deploying ML, DL and Generative AI applications.

TECHNICAL SKILLS

Programming: Python, SQL, Data Structures and Algorithms (DSA).
Machine Learning: Supervised & Unsupervised Learning, Ensemble Learning, Model Evaluation, Regularization (Lasso, Ridge), Overfitting & Underfitting**.**
Algorithms: Linear Regression, Logistic Regression, Decision Trees, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naive Bayes, K-Means, DBSCAN, Principal Component Analysis (PCA), Hierarchical Clustering, Isolation Forest, Local Outlier Factor (LOF)**.**
Deep Learning: Artificial Neural Networks (ANN), Feedforward Neural Networks (FNN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Transformers.
Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Generative Adversarial Networks (GANs), OpenAI API, Natural Language Processing (NLP).
Libraries and Frameworks: scikit-learn, PyTorch, NumPy, Pandas, Matplotlib, Seaborn, LangChain, Streamlit, FastAPI.
Cloud & DevOps: AWS, Docker, CI/CD, Git/GitHub

PROJECTS

AI-Powered Cyber Threat Detection System

- Developed a binary network intrusion detection pipeline using Logistic Regression and XGBoost, preprocessing eight network-traffic datasets and selecting 60 features.
- Evaluated the model on an unseen Friday traffic dataset of 516,824 network-flow records, reporting 99.81% accuracy; the evaluation recorded 99.97% overall attack recall.
- Deployed a Streamlit application for CSV uploads, attack-probability estimates, batch predictions, detection summaries, and downloadable results.

Technologies: Python, Pandas, NumPy, scikit-learn, XGBoost, Streamlit, Joblib
GitHub: GitHub Repository | Live Demo: View Demo

AI-Powered Document Intelligence (RAG)

- Built an end-to-end PDF question-answering application that generates context-grounded responses using a large language model.
- Implemented PDF extraction, text chunking, Hugging Face embeddings, FAISS semantic retrieval, and source-page references.
- Deployed a Streamlit interface supporting multiple PDFs, conversational Q&A, chat history, and document statistics.

Technologies: Python, Streamlit, LangChain, Groq LLM, FAISS, Hugging Face Embeddings, PyPDF
GitHub: GitHub Repository | Live Demo: View Demo

AI YouTube Video Analyzer

- Developed an AI-powered video analysis application using an Agno AI Agent and Groq LLM to generate summaries, study notes, quizzes, key takeaways, and timestamp-based insights.
- Built a Streamlit interface with YouTube metadata extraction and downloadable PDF and text reports.

Technologies: Python, Streamlit, Agno AI, Groq LLM, yt-dlp, ReportLab.
GitHub: GitHub Repository | Live Demo: View Demo

EDUCATION

B.E. in Artificial Intelligence and Machine Learning | ACET College | CGPA: 7.63/10 | Expected Graduation: July 2027
Class XII (DPUE) | Govt. Ex-Municipal PU College | 86.5% | 2023

CERTIFICATION

Prime (AI/ML) — Certificate of Completion, Apna College | Certificate: [View Certificate]